Finite Regional Residue Obstructions Under Gluing: Associator Fields, Seam Repair, and Computable Cohomology Witnesses

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Abstract

Local algebraic operations may be associative on every region while boundary-corrected seam products fail to associate after regional data are glued. This paper studies the finite datum produced by that failure before imposing a cohomological, spectral, or verification-theoretic classification on it. The primary object is an associator field: an assignment of a boundary-supported defect to each non-empty triple overlap of a finite regional cover.

Regions are taken to form a finite distributive lattice. Algebraic data are assigned to regions by injective extension maps, a support-refining multiplication rule, and boundary corrections supported on pairwise overlaps. The first main theorem proves that the associator defect of a corrected triple product is forced onto the triple overlap. Thus gluing creates a diagnostic level between pairwise compatibility and global coherence: every single region may be internally associative, and every pairwise seam may be explicitly corrected, while the threefold composition still leaves a residue at the common seam.

The paper then separates three questions that are often conflated. Detection asks where the associator field is non-zero. Repair asks whether it can be removed by admissible changes of pairwise boundary bookkeeping. Obstruction asks whether a persistent component remains after all such repairs. Ordinary Čech cohomology is recovered only under additional restriction, alternation, and coherence hypotheses. Spectral Hodge analysis is available when the defect field is scalar or finite-dimensional over an inner-product field.

The final part of the paper gives a computable finite obstruction witness. A four-region seam residue over $\mathbb{Q}$ is shown to be a cocycle but not a coboundary, hence to define a non-zero class in H^1. The proof is given twice: by an explicit linear contradiction and by a non-zero cycle pairing. The same obstruction verdict is invariant under declared presentation changes and persists under four declared refinement witnesses. The resulting claim is deliberately scoped: the paper proves the stated finite obstruction and its tested refinement witnesses, while identifying the additional pushforward-pullback pairing theorem required for a universal refinement result.

1 Introduction

Associativity is usually presented as an internal algebraic law. An algebra is associative if

$$(ab)c = a(bc)$$

for all elements. That formulation is correct, but it hides a boundary problem. If an algebraic operation is assembled from local data on overlapping regions, then there are two separate questions. First, does multiplication associate inside each region? Second, does multiplication continue to associate after local products have been glued across seams?

The second question is not a restatement of the first. A gluing system may have associative algebras on every local region. It may also have explicit correction terms on every pairwise overlap.

Nevertheless, threefold composition may depend on parenthesisation. Schematically, the defect has the form

$$a \star (b \star c) - (a \star b) \star c.$$

With types restored, this defect is not an interior failure of any one algebra. It is a residue of the gluing process itself.

This distinction is familiar in computational form. A module may satisfy its local specification. Two modules may satisfy a pairwise interface contract. A third module may still expose an integration failure that was invisible to the pairwise checks. Likewise, three sensor regions may maintain interval estimates of a shared quantity. Each pairwise reconciliation may be well-defined, but the two histories of threefold reconciliation may disagree at the common seam. This paper does not reduce the mathematics to these applications. It uses them to make the boundary problem visible.

The paper studies that residue directly. The point is not, at first, to produce a Čech class. The point is to identify the finite field of defects generated by boundary-corrected composition. For a cover $\mathcal{U} = (U_i)_{i \in I}$, the datum has the form

$$(i, j, k) \longmapsto \alpha_{ijk}.$$

Here α_{ijk} is the associator defect associated with the triple (U_i, U_j, U_k). Under the support hypotheses stated below, this defect is supported on

$$U_i \wedge U_j \wedge U_k.$$

The failure is therefore localised, but it is not merely local. It is produced by comparing two histories of composition across a larger gluing pattern.

This distinction is the organising idea of the paper. Gluing creates a diagnostic level between pairwise compatibility and global coherence. A system can pass all unary checks, since each $A(U_i)$ is internally associative. It can pass all binary checks, since each pairwise overlap $U_i \wedge U_j$ carries an explicit boundary correction. It can still fail a ternary check, since the two ways of composing across U_i, U_j, U_k need not agree. The associator field is the finite record of that third level.

The construction is deliberately austere. It does not try to characterise every possible failure of algebraic descent. It does not begin from the full machinery of sheaves, cosheaves, factorisation algebras, or higher categories. It isolates a small finite setting in which the phenomenon can be seen without analytic overhead: a finite point-free system of regions, local algebras with support information, and boundary corrections living on pairwise overlaps. This restriction is methodological. It keeps the phenomenon visible.

The new element of the present version is that the diagnostic separation is carried through to an explicit finite obstruction witness. The paper first builds the associator-field machinery, then shows how a seam-supported residue may be classified by exact rational cohomology. This matters because a non-zero residue is not automatically an obstruction. It may be removable by changing local representatives or pairwise bookkeeping. The finite classifier asks the decisive question:

$$r \in \ker \delta^1 \qquad \text{and} \qquad r \notin \operatorname{im} \delta^0?$$

When both conditions hold, the residue defines a non-zero class

$$0 \neq [r] \in H^1.$$

1.1 Main claims

The paper establishes the following claims.

(1) **Localisation.** In a supported regional algebra with support-refining boundary corrections, the associator defect of a corrected triple product is supported on the triple overlap.

(2) **Associator field.** For a finite cover and a family of local inputs, the localised defects assemble into a finite field indexed by the non-empty triple overlaps of the nerve.

(3) **Repair before obstruction.** The first classification of the field is not cohomological. It is operational: which part of the field is removed by admissible changes of pairwise boundary bookkeeping?

(4) **Cohomology as a special case.** If a genuine overlap coefficient system with restriction maps is supplied, if an alternating realisation of the associator field is chosen, and if the realised field satisfies the relevant fourfold coherence condition, then the field determines a Čech-type obstruction class.

(5) **Finite residue obstruction.** In the four-region finite example of Sections 11 and 12, the seam residue $r = (1, 1, 1, -2)$ over $\mathbb{Q}$ is a cocycle but not a coboundary. Hence it represents a non-zero class in H^1.

(6) **Witnessed persistence.** The obstruction verdict is invariant under the declared presentation changes and persists under four declared refinement witnesses. This is a proved finite statement about the displayed witnesses, not a universal theorem about all refinements.

The order matters. The associator field comes first. Cohomological and spectral classifications enter only after the field has been constructed and its support behaviour understood. The finite obstruction witness then shows how one such classification can be made exact and reproducible.

1.2 Running example and reader's guide

A single example will recur informally before being computed in Section 9. Suppose several nodes maintain interval views of a shared state. Their overlaps record shared interfaces. Pairwise boundary corrections record reconciliation costs at those interfaces. The central question is not whether each node can multiply or update its own data associatively. Nor is it only whether every pair of nodes can reconcile. The question is whether the threefold history of reconciliation is independent of parenthesisation.

This running example should not be confused with the general definition. It is an interpretive aid, not a premise of the theory. The formal theory below works with finite distributive lattices, supported algebras, and bilinear boundary corrections. The interval model merely shows why the definitions have the shape they do: support tracks where a contribution lives, boundary correction tracks pairwise seam bookkeeping, and the associator field records the residue left at a triple seam.

1.3 Relation to existing work

Mac Lane's coherence theorem shows that associativity constraints in a monoidal category behave coherently once the relevant associators are supplied as natural isomorphisms [6, 7]. The present paper studies a different problem. It asks how associators arise as defects when multiplication is glued from regional data.

The local-to-global pattern resembles descent theory. Classical descent asks whether local objects and compatibility data determine a global object [4, 8]. Here the compatibility datum is not an isomorphism between pullbacks. It is a boundary-corrected multiplication. The first object of study is therefore not a descent datum, but the associator residue produced when corrected binary gluing is iterated.

The paper is also adjacent to sheaf and cosheaf methods [5]. It remains below their full generality. Rather than assuming exact gluing axioms, it asks what algebraic residue appears when gluing fails to preserve associativity. In that sense, the paper occupies an intermediate level: more structured than an informal interface-failure analogy, but less general than a full descent or factorisation-algebra framework [2].

2 Region systems and covers

Definition 2.1 (Region system)**.** *A region system is a finite distributive lattice*

$$(\mathrm{Reg}, \leq, \perp, \top, \vee, \wedge).$$

The elements of Reg *are called regions. The join* $\vee$ *is read as regional union, and the meet* $\wedge$ *as overlap. No underlying point-set representation is assumed.*

Definition 2.2 (Finite cover)**.** *Let* $R \in \mathrm{Reg}$*. A finite family* $\mathcal{U} = (U_i)_{i \in I}$ *is a cover of* R *if*

$$R = \bigvee_{i \in I} U_i.$$

For every non-empty finite subset $J \subseteq I$*, write*

$$U_J = \bigwedge_{j \in J} U_j.$$

For $J = \{i_0, \ldots, i_n\}$*, we also write* $U_{i_0 \cdots i_n}$*.*

Definition 2.3 (Nerve)**.** *The nerve* $N(\mathcal{U})$ *is the finite simplicial complex whose* n*-simplices are the non-empty subsets* $J \subseteq I$ *with* $|J| = n + 1$ *and*

$$U_J \neq \perp.$$

Thus vertices are regions, edges are non-empty pairwise overlaps, and triangles are non-empty triple overlaps.

Remark 2.4 (Why the nerve is not yet the answer)**.** *The nerve records the incidence pattern of overlaps. It does not by itself say what lives on those overlaps. The associator field constructed below uses the nerve as an index shape, but its values come from the regional algebra and its boundary corrections.*

Definition 2.5 (Ordered simplices)**.** *When an ordered tuple* $(i_0, \ldots, i_n)$ *is used, its entries are assumed to be pairwise distinct unless explicitly stated otherwise. Such a tuple represents an oriented simplex of the nerve whenever* $U_{i_0 \cdots i_n} \neq \perp$*.*

3 Supported regional algebras

The localisation theorem needs a disciplined notion of support. The next definition supplies it. The algebras are not required to be unital, since extension by zero is one intended model.

Definition 3.1 (Supported regional algebra)**.** *Let k be a commutative ring. A supported regional algebra on* Reg *consists of the following data.*

(i) For each region $R \in \mathrm{Reg}$, a not-necessarily-unital associative k-algebra $A(R)$.

(ii) For each inclusion $R \leq S$, an injective k-algebra homomorphism

$$\iota_{RS} : A(R) \to A(S),$$

not required to preserve units, satisfying

$$\iota_{RR} = \mathrm{id}, \qquad \iota_{ST} \circ \iota_{RS} = \iota_{RT} \quad (R \leq S \leq T).$$

(iii) An intersection support-refinement rule: if $x \in A(S)$ is supported on $R \leq S$, and $y \in A(S)$ is supported on $T \leq S$, then xy is supported on $R \wedge T$.

Here an element $x \in A(S)$ is supported on $R \leq S$ when

$$x \in \mathrm{im}(\iota_{RS}).$$

Because ι_{RS} is injective, such an element has a unique representative in $A(R)$, denoted $x|_R$. No minimal support is assumed: an element may be supported on several regions, and $x|_R$ always refers to the representative relative to a specified supporting region R, not to a smallest such region.

Remark 3.2 (Running example)**.** *In the interval reconciliation model of Section 9, the primary component records an ordinary interval value or width, while a square-zero annotation channel records seam cost. Support says where that seam cost lives. The point of the support rule is to prevent a boundary contribution produced at one overlap from being silently treated as an interior contribution somewhere else.*

Remark 3.3 (Extension is not restriction)**.** *The maps ι_{RS} go from smaller regions to larger regions. They are extension maps. The notation $x|_R$ is used only when x is already known to lie in the image of ι_{RS}. It is not a general restriction map $A(S) \to A(R)$.*

Lemma 3.4 (Supported multiplication representative)**.** *Let $R, T \leq S$. If $x \in A(S)$ is supported on R, and $y \in A(S)$ is supported on T, then xy has a unique representative*

$$(xy)|_{R\wedge T} \in A(R \wedge T).$$

Proof. By the support-refinement rule, xy is supported on $R \wedge T$. Hence $xy \in \mathrm{im}(\iota_{R\wedge T,S})$. Since $\iota_{R\wedge T,S}$ is injective, the representative is unique. □

4 Boundary corrections and corrected products

A boundary correction records a contribution assigned to the seam where two regions are glued.

Definition 4.1 (Boundary correction). *A boundary correction on a supported regional algebra is a family of k-bilinear maps*

$$\beta_{U,V} : A(U) \otimes_k A(V) \to A(U \wedge V)$$

for ordered pairs of regions (U, V).

Definition 4.2 (Support-refining boundary correction). *A boundary correction β is support-refining if the following holds. Let $X \le U$ and $Y \le V$. If $a \in A(U)$ is supported on X, and $b \in A(V)$ is supported on Y, then $\beta_{U,V}(a, b) \in A(U \wedge V)$ is supported on*

$$X \wedge Y \le U \wedge V.$$

Remark 4.3 (Running example). *For interval reconciliation, a boundary correction can be as simple as an on-off seam annotation. One may set*

$$\beta_{X,Y}(a, b) = \chi(X, Y)\pi(a)\pi(b)\varepsilon,$$

where $\chi(X, Y) = 1$ when two interval views properly partially overlap and $\chi(X, Y) = 0$ otherwise. The general definition allows more than this scalar toy model, but the toy model explains why β is bilinear and why its value belongs to the overlap algebra.

Definition 4.4 (Raw overlap product). *For $a \in A(U)$ and $b \in A(V)$, set $S = U \vee V$. Extend both elements to S, multiply there, and take the supported representative on $U \wedge V$:*

$$m_{U,V}(a, b) := \big(\iota_{US}(a)\iota_{VS}(b)\big)|_{U\wedge V} \in A(U \wedge V).$$

This is well-defined by Lemma 3.4.

Definition 4.5 (Boundary-corrected product). *For $a \in A(U)$ and $b \in A(V)$, define*

$$a \star_{U,V} b := \iota_{U\wedge V,U\vee V}\big(m_{U,V}(a, b) + \beta_{U,V}(a, b)\big) \in A(U \vee V).$$

Remark 4.6 (Boundary interaction product). *The product $\star_{U,V}$ is a boundary interaction product. Its value lies in the ambient algebra $A(U \vee V)$, but its support is contained in $U \wedge V$. This is intentional. The paper isolates the seam contribution rather than modelling arbitrary independent interior multiplication on $U \vee V$.*

Lemma 4.7 (Support of the corrected product). *For $a \in A(U)$ and $b \in A(V)$, the element*

$$a \star_{U,V} b \in A(U \vee V)$$

is supported on $U \wedge V$.

Proof. Both $m_{U,V}(a, b)$ and $\beta_{U,V}(a, b)$ lie in $A(U \wedge V)$. The corrected product is obtained by applying $\iota_{U\wedge V,U\vee V}$ to their sum. □

5 The associator defect

Let $U, V, W \in \mathrm{Reg}$, and let

$$a \in A(U), \qquad b \in A(V), \qquad c \in A(W).$$

There are two corrected ways to multiply these three elements:

$$(a \star_{U,V} b) \star_{U \vee V, W} c, \qquad a \star_{U, V \vee W} (b \star_{V,W} c).$$

Both lie in $A(U \vee V \vee W)$.

Definition 5.1 (Associator defect)**.** *The associator defect of a, b, c over (U, V, W) is*

$$\alpha_{U,V,W}(a, b, c) := a \star_{U,V \vee W} (b \star_{V,W} c) - (a \star_{U,V} b) \star_{U \vee V,W} c \in A(U \vee V \vee W).$$

This is the right-minus-left convention used throughout the paper.

Lemma 5.2 (Extension preserves support)**.** *Let $R \leq T \leq S$. If $x \in A(T)$ is supported on R, then $\iota_{TS}(x) \in A(S)$ is supported on R.*

Proof. Since x is supported on R, there is $x_R \in A(R)$ such that $x = \iota_{RT}(x_R)$. Hence

$$\iota_{TS}(x) = \iota_{TS}\iota_{RT}(x_R) = \iota_{RS}(x_R),$$

so $\iota_{TS}(x) \in \mathrm{im}(\iota_{RS})$. □

Theorem 5.3 (Triple-overlap localisation)**.** *Assume that β is support-refining. Then*

$$\alpha_{U,V,W}(a, b, c)$$

is supported on

$$U \wedge V \wedge W.$$

Equivalently, there is a unique element

$$\widetilde{\alpha}_{U,V,W}(a, b, c) \in A(U \wedge V \wedge W)$$

such that

$$\iota_{U \wedge V \wedge W, U \vee V \vee W}\big(\widetilde{\alpha}_{U,V,W}(a, b, c)\big) = \alpha_{U,V,W}(a, b, c).$$

Proof. Set $S = U \vee V \vee W$. Consider first the left-parenthesised product

$$(a \star_{U,V} b) \star_{U \vee V,W} c.$$

By Lemma 4.7, the inner product $a \star_{U,V} b$ is supported on $U \wedge V$ inside $A(U \vee V)$. By Lemma 5.2, after extension to S it remains supported on $U \wedge V$. The element c, extended from W to S, is supported on W. By the intersection support-refinement rule, the raw outer product of these two elements is supported on

$$(U \wedge V) \wedge W = U \wedge V \wedge W.$$

The outer correction $\beta_{U \vee V,W}(a \star_{U,V} b, c)$ is supported on $(U \wedge V) \wedge W = U \wedge V \wedge W$ as well, by support refinement for β applied to the first argument supported on $U \wedge V$ and the second supported on W. Hence the left-parenthesised product is supported on $U \wedge V \wedge W$.

The right-parenthesised product

$$a \star_{U,V\vee W} (b \star_{V,W} c)$$

is treated by the same argument with the roles of the union terms exchanged. By Lemma 4.7, the inner product $b \star_{V,W} c$ is supported on $V \wedge W$, and by Lemma 5.2 it remains so after extension to S. The element a is supported on U. The raw outer product is therefore supported on $U \wedge (V \wedge W)$, and the outer correction $\beta_{U,V\vee W}(a, b \star_{V,W} c)$ is supported on the same overlap by support refinement. Since the meet is associative,

$$U \wedge (V \wedge W) = U \wedge V \wedge W,$$

so the right-parenthesised product is supported on $U \wedge V \wedge W$ as well.

Both parenthesised products are therefore supported on $U \wedge V \wedge W$, and so is their difference. The uniqueness of the representative $\widetilde{\alpha}_{U,V,W}(a, b, c)$ follows from injectivity of the extension map $\iota_{U\wedge V\wedge W,S}$. □

Remark 5.4 (Localisation, not mere failure). *Theorem 5.3 says more than that associativity can fail. It says that, under explicit support discipline, the failure is forced onto the common seam of the three regions. The defect is locally supported but globally induced.*

6 Associator fields

We now pass from one triple to a finite cover.

Definition 6.1 (Input family). *Let $\mathcal{U} = (U_i)_{i\in I}$ be a finite cover. An input family on $\mathcal{U}$ is a choice*

$$a = (a_i)_{i\in I}, \qquad a_i \in A(U_i).$$

Definition 6.2 (Associator field). *Let $\mathcal{U} = (U_i)_{i\in I}$ be a finite cover and let $a = (a_i)_{i\in I}$ be an input family. The associator field of a is the assignment*

$$\mathcal{A}_a : (i, j, k) \longmapsto \widetilde{\alpha}_{U_i,U_j,U_k}(a_i, a_j, a_k) \in A(U_i \wedge U_j \wedge U_k)$$

for ordered triples of pairwise distinct indices with $U_i \wedge U_j \wedge U_k \neq \perp$.

Remark 6.3 (The field is ordered, not alternating). *No antisymmetry in (i, j, k) is assumed. The field is ordered because the corrected product need not be symmetric in its inputs, so in general*

$$\mathcal{A}_a(i, j, k) \neq -\mathcal{A}_a(j, i, k).$$

This matters later: passing to a Čech cochain requires an explicit alternation hypothesis, rather than following automatically.

Remark 6.4 (Field before class). *The associator field is not automatically a cocycle, a cohomology class, or a harmonic form. It is the finite datum produced by gluing. Further structures may classify it, but they do not constitute it.*

The field has three primitive features.

(i) **Incidence:** it is defined only where the triple overlap exists.

(ii) **Support:** each value lives on the corresponding triple overlap.

(iii) **History:** each value compares two parenthesised histories of composition.

This gives the first conceptual separation:

$$\text{interior validity} \neq \text{pairwise compatibility} \neq \text{triple coherence}.$$

Interior validity means that each $A(U_i)$ is associative. Pairwise compatibility means that each seam $U_i \wedge U_j$ carries a specified correction. Triple coherence means that the corrections do not leave an associator residue on $U_i \wedge U_j \wedge U_k$.

7 Detection, repair, obstruction

The associator field invites three different questions. They should be kept separate.

7.1 Detection

Detection asks whether and where the field is non-zero:

$$\mathcal{A}_a(i, j, k) \neq 0.$$

This is the most primitive diagnostic. It identifies triple seams at which pairwise-corrected composition still depends on parenthesisation.

7.2 Repair

Repair asks whether the defect can be removed by changing pairwise boundary bookkeeping.

Definition 7.1 (Boundary gauge, informal finite version). *A boundary gauge is an admissible change*

$$\beta_{U,V} \longmapsto \beta_{U,V} + \Theta_{U,V},$$

where each

$$\Theta_{U,V} : A(U) \otimes_k A(V) \to A(U \wedge V)$$

is bilinear and support-refining.

Definition 7.2 (Admissible gauge in the first-order regime). *In the first-order boundary regime of Section 8, a gauge Θ is admissible if each $\Theta_{U,V}$ is bilinear and support-refining, its outputs lie in the relevant boundary ideal after extension to the ambient region, and it vanishes whenever either argument lies in the relevant boundary ideal. The informal use of admissibility above is made precise by this condition once the first-order regime has been fixed.*

The word admissible is doing real work. A repair is not an arbitrary assignment to triple overlaps. It must be induced by changes at pairwise seams. Thus repair is a constrained problem: can pairwise changes cancel a triple-seam field?

In a linearised or first-order regime, boundary gauges determine a variation map

$$D : G^1 \to \mathrm{Field}^2,$$

from admissible pairwise gauges to associator fields. Here G^1 denotes the k-module of admissible pairwise gauge families (Θ_{ij}), and Field^2 denotes the k-module of triple-indexed assignments

$$F_{ijk} \in A(U_i \wedge U_j \wedge U_k),$$

both for the fixed cover and input family under discussion. A field $\mathcal{A}$ is repairable when

$$\mathcal{A} \in \operatorname{im} D.$$

Definition 7.3 (Repair quotient). *In a fixed first-order regime with variation map $D : G^1 \to \mathrm{Field}^2$, define the repair quotient*

$$\mathrm{Ob}_D := \mathrm{Field}^2 \,/\, \mathrm{im}\, D.$$

The class of $\mathcal{A}$ in Ob_D records the part of the associator field that is not removed by admissible pairwise boundary gauges.

Remark 7.4 (Why this is not yet Čech cohomology). *The operator D need not be the ordinary Čech coboundary. It is determined by how the corrected product is parenthesised and by which boundary gauges are admissible. Ordinary Čech cohomology appears only when D can be identified with a Čech differential in a specified coefficient system.*

7.3 Obstruction

Obstruction is the modal status of a defect after repair has been attempted. A non-zero value

$$\mathcal{A}_a(i, j, k) \neq 0$$

is not yet an obstruction. It may be removable by a change of pairwise bookkeeping. It becomes obstruction-like only when its class in the repair quotient is non-zero:

$$[\mathcal{A}_a]_D \neq 0.$$

Thus obstruction is not the first object. It is what remains when admissible repairs fail.

8 First-order boundary variation

This section records the finite first-order shape of repair. It is deliberately stated as a linearised calculation rather than as a universal theorem about all boundary-corrected products.

Definition 8.1 (First-order boundary regime). *A boundary-corrected regional algebra is first-order if, for every region S, there is a two-sided k-submodule ideal $B(S) \subseteq A(S)$, called the boundary ideal, such that:*

(i) for every pair U, V and all inputs a, b, the extended correction $\iota_{U\wedge V,U\vee V}(\beta_{U,V}(a, b))$ and the extended gauge $\iota_{U\wedge V,U\vee V}(\Theta_{U,V}(a, b))$ lie in $B(U \vee V)$;

(ii) $B(S)^2 = 0$;

(iii) multiplication by ordinary elements sends boundary elements to boundary elements;

(iv) extension maps preserve the boundary ideals;

(v) for every pair U, V, the maps $\beta_{U,V}$ and every admissible gauge $\Theta_{U,V}$ factor through

$$(A(U)/B(U)) \otimes_k (A(V)/B(V)),$$

equivalently they vanish whenever either input lies in the relevant boundary ideal.

In such a regime, products of two newly inserted boundary corrections vanish. The effect of changing β by a gauge Θ is therefore linear to first order.

Remark 8.2 (Running example). *The dual-number model $k[\varepsilon]/(\varepsilon^2)$ is the simplest first-order regime. The ordinary channel is the coefficient of* 1. *The boundary channel is the coefficient of* ε. *Since $\varepsilon^2 = 0$, two newly inserted reconciliation annotations do not multiply into a second-order ordinary effect.*

Proposition 8.3 (Four-term first-order variation). *Let A be a first-order boundary regime, let $a \in A(U)$, $b \in A(V)$, $c \in A(W)$, write $\Lambda = U \wedge V \wedge W$, and let $m_{X,Y}$ denote the uncorrected raw overlap product. If the boundary correction β is changed by an admissible first-order gauge Θ, then the associator defect, with the right-minus-left convention, changes by $D\Theta_{U,V,W}(a,b,c)$, the unique element of $A(\Lambda)$ given by*

$$\begin{aligned}
D\Theta_{U,V,W}(a,b,c) = {} & \Big[\Theta_{U,V\vee W}\big(a, \iota_{V\wedge W,V\vee W}(m_{V,W}(b,c))\big)\Big]|_\Lambda \\
& + m_{U,V\wedge W}\big(a, \Theta_{V,W}(b,c)\big) \\
& - \Big[\Theta_{U\vee V,W}\big(\iota_{U\wedge V,U\vee V}(m_{U,V}(a,b)), c\big)\Big]|_\Lambda \\
& - m_{U\wedge V,W}\big(\Theta_{U,V}(a,b), c\big).
\end{aligned}$$

Proof. Expand the two parenthesised products after replacing β by $\beta + \Theta$. Because the regime is first-order, every term carrying two newly inserted boundary corrections vanishes, and the admissible gauge maps ignore arguments that are already boundary elements. Hence the variation is linear in Θ.

The right-parenthesised product contributes two first-order terms: the outer gauge evaluated on the raw inner product, after the inner product is extended from $V \wedge W$ to $V \vee W$, and the raw outer product evaluated on the inner gauge. The first term initially lands in a larger overlap algebra and is reduced to Λ by $|_\Lambda$. These are the first two displayed terms.

The left-parenthesised product contributes the corresponding two terms with negative sign, because the associator defect is right-minus-left. Every displayed map has a determined source and target. The bracketed terms require the support witnesses that identify their outputs with elements supported on Λ. With those witnesses, support refinement places every displayed term on Λ, and injectivity of the extension maps gives the stated representatives in $A(\Lambda)$. □

Remark 8.4 (The warning hidden in the formula). *The four-term expression is the native repair operator for this parenthesised gluing problem. It is not automatically the ordinary three-face Čech formula. To obtain ordinary Čech cohomology, one must impose additional identifications that replace the mixed union terms by the corresponding face restrictions in a genuine coefficient system.*

9 A first-order interval example

This section gives a concrete scalar instance of the first-order calculation. It first records a scalar dual-number seam model and then specialises that model to interval reconciliation.

Let all local algebras be the dual-number algebra

$$D = k[\varepsilon]/(\varepsilon^2).$$

Let the boundary ideal be $k\varepsilon$. Products of two boundary corrections vanish. For this calculation, assume that all extension maps are identities and that U, V, W have non-empty triple overlap. This is a scalar coefficient reduction of the regional theory, not a faithful model of minimal support. With all extension maps taken as identities, every element is supported on every region, so the calculation computes the seam coefficients while suppressing the proof-carrying support representatives of the general theory.

For constants $\lambda_{P,Q} \in k$, define

$$\beta_{P,Q}(x, y) = \lambda_{P,Q}\pi(x)\pi(y)\varepsilon,$$

where

$$\pi(x_0 + x_1\varepsilon) = x_0.$$

Thus the correction depends only on the primary channel and vanishes if either input is purely boundary. For

$$a = a_0 + a_1\varepsilon, \qquad b = b_0 + b_1\varepsilon, \qquad c = c_0 + c_1\varepsilon,$$

a direct expansion gives

$$\widetilde{\alpha}_{U,V,W}(a, b, c) = \Delta_{U,V,W}\, a_0 b_0 c_0 \varepsilon,$$

where

$$\Delta_{U,V,W} = \lambda_{V,W} - \lambda_{U\vee V,W} + \lambda_{U,V\vee W} - \lambda_{U,V}.$$

This is the native first-order boundary expression for the parenthesised gluing problem. If this expression is induced by an admissible pairwise gauge, the defect is repairable. If no admissible gauge induces it, the defect persists in the repair quotient.

Example 9.1 (Annotation residue of interval reconciliation)**.** *We instantiate the model as a distributed system whose nodes maintain interval estimates of a shared state, for instance sensor ranges, clock-synchronisation windows, or bounding boxes. The primary channel of $D = k[\varepsilon]/(\varepsilon^2)$ carries an interval width, and the boundary ideal $k\varepsilon$ records the square-zero reconciliation cost incurred whenever two regions must actively reconcile shared state.*

For regions presented as intervals, define the reconciliation indicator

$$\chi(X, Y) = \begin{cases} 1, & X \cap Y \neq \varnothing \text{ and neither } X \subseteq Y \text{ nor } Y \subseteq X, \\ 0, & \text{otherwise.} \end{cases}$$

and take

$$\beta_{X,Y}(a, b) = \chi(X, Y)\pi(a)\pi(b)\varepsilon.$$

This is the scalar first-order seam model with $\lambda_{X,Y} = \chi(X, Y)$.

Consider three nodes holding interval projections of one variable, with local views

$$U = [0, 4], \qquad V = [2, 6], \qquad W = [4, 8].$$

The pairwise interfaces are

$$U \wedge V = [2, 4], \qquad V \wedge W = [4, 6], \qquad U \wedge W = \{4\},$$

and the triple consensus seam is

$$U \wedge V \wedge W = \{4\}.$$

Evaluating the field on the constant weights $a = b = c = 1$, the four binary gluing sites give

$$\chi(U, V) = 1, \quad \chi(V, W) = 1, \quad \chi(U \vee V, W) = 1, \quad \chi(U, V \vee W) = 1.$$

Hence

$$\widetilde{\alpha}_{U,V,W}(1, 1, 1) = (1 - 1 + 1 - 1)\varepsilon = 0.$$

In this configuration, both parenthesisations incur the same first-order reconciliation residue, and the associator field vanishes.

Now reposition the third node so that its projection falls inside the combined span of the first two nodes while still crossing their interface:

$$U = [0,4], \qquad V = [2,6], \qquad W = [1,5].$$

Then $W \subseteq U \vee V = [0,6]$*, but* W *is not contained in* U *or* V *individually. The indicators are*

$$\chi(U,V) = 1, \quad \chi(V,W) = 1, \quad \chi(U, V \vee W) = 1, \quad \chi(U \vee V, W) = 0.$$

The common triple seam is

$$U \wedge V \wedge W = [2,4].$$

Therefore

$$\widetilde{\alpha}_{U,V,W}(1,1,1) = (1 - 0 + 1 - 1)\varepsilon = \varepsilon \neq 0.$$

Thus the non-zero residue is not produced by an interior failure of any local algebra. Nor is it produced by an empty or degenerate overlap. It is produced at the triple seam $[2,4]$*, where two parenthesised histories of interval reconciliation disagree.*

The lesson is precise: pairwise reconciliation does not by itself guarantee triple coherence. The defect appears exactly when a region is absorbed by a union while still crossing the interface between the union's components.

10 When the field becomes Čech data

We now state the additional structure under which the associator field may be interpreted cohomologically. This is a specialisation, not the starting point.

Definition 10.1 (Overlap coefficient system)**.** *Let* $\mathcal{U} = (U_i)_{i \in I}$ *be a finite cover. An overlap coefficient system assigns an abelian group* $E(U_J)$ *to each non-empty overlap* U_J*, together with restriction maps*

$$\rho_{KJ} : E(U_K) \to E(U_J) \qquad (K \subseteq J),$$

satisfying

$$\rho_{JJ} = \mathrm{id}, \qquad \rho_{JL} \circ \rho_{KJ} = \rho_{KL} \quad (K \subseteq J \subseteq L).$$

Remark 10.2 (Do not confuse the two variances)**.** *The algebraic assignment* A *has extension maps from smaller regions to larger regions. A Čech coefficient system has restriction maps from larger overlaps to smaller overlaps. These are different structures. One may take* $E(U_J) = A(U_J)$ *only if suitable restriction maps have actually been supplied.*

Definition 10.3 (Čech cochains on the nerve)**.** *Let* E *be an overlap coefficient system on* $\mathcal{U}$*. For* $n \geq 0$*, define* $\check{C}^n(\mathcal{U}; E)$ *to be the abelian group of alternating assignments*

$$c_{i_0 \cdots i_n} \in E(U_{i_0 \cdots i_n})$$

for ordered $(n+1)$*-tuples of pairwise distinct indices with* $U_{i_0 \cdots i_n} \neq \perp$*. The coboundary*

$$\delta : \check{C}^n(\mathcal{U}; E) \to \check{C}^{n+1}(\mathcal{U}; E)$$

is given by

$$(\delta c)_{i_0 \cdots i_{n+1}} = \sum_{r=0}^{n+1} (-1)^r \rho_{I \setminus \{i_r\}, I}\big(c_{i_0 \cdots \widehat{i_r} \cdots i_{n+1}}\big),$$

where $I = \{i_0, \ldots, i_{n+1}\}$*.*

Definition 10.4 (Coefficient realisation of the associator field). *A coefficient realisation of the associator field is a family of maps*

$$\lambda_J : A(U_J) \to E(U_J)$$

for non-empty overlaps. No restriction structure on A is assumed; the restriction structure needed for Čech theory is carried by E. Given an input family a, it produces the ordered assignment

$$(\lambda \mathcal{A}_a)_{ijk} := \lambda_{ijk}\big(\mathcal{A}_a(i,j,k)\big) \in E(U_{ijk})$$

on ordered triples of pairwise distinct indices with $U_{ijk} \neq \perp$.

Definition 10.5 (Alternating coefficient realisation). *A coefficient realisation λ is alternating for the input family a if the ordered assignment satisfies*

$$(\lambda \mathcal{A}_a)_{i_{\sigma(0)} i_{\sigma(1)} i_{\sigma(2)}} = \mathrm{sgn}(\sigma)(\lambda \mathcal{A}_a)_{i_0 i_1 i_2}$$

for every permutation σ of $\{0,1,2\}$. In that case $\lambda \mathcal{A}_a \in \check{C}^2(\mathcal{U}; E)$.

Definition 10.6 (Fourfold coherence). *The realised associator field $\lambda \mathcal{A}_a$ is fourfold coherent if*

$$\delta(\lambda \mathcal{A}_a) = 0$$

on every non-empty quadruple overlap.

Proposition 10.7 (Čech class under extra hypotheses). *If an overlap coefficient system E, a coefficient realisation λ that is alternating for a, and fourfold coherence are given, then the realised associator field defines a class*

$$[\lambda \mathcal{A}_a] \in \check{H}^2(\mathcal{U}; E).$$

Proof. Since λ is alternating for a, the ordered assignment $\lambda \mathcal{A}_a$ is a genuine 2-cochain in $\check{C}^2(\mathcal{U}; E)$. Fourfold coherence says that its Čech coboundary vanishes. Hence it is a cocycle, and therefore determines a cohomology class. □

Remark 10.8 (What the proposition does not say). *The proposition does not say that every associator field is automatically a Čech cocycle. It says that a realised associator field becomes one after the correct restriction structure, alternation convention, and fourfold coherence condition have been supplied.*

11 A finite residue classifier

The preceding sections concern associator fields on triple seams. This section isolates a related but lower-degree question: once a finite seam residue has been produced, how do we decide whether it is removable or genuinely obstructive?

The answer is ordinary finite cohomology over an exact coefficient field. The important point is conceptual, not technical. A non-zero residue is not yet an obstruction. It becomes an obstruction only when it is closed and not exact.

Let

$$C^0 \xrightarrow{\delta^0} C^1 \xrightarrow{\delta^1} C^2$$

be a finite cochain complex over $\mathbb{Q}$. A degree-one residue is an element $r \in C^1$. It has three possible statuses.

(1) If $\delta^1 r \neq 0$, the residue is a coherence failure. It is not a degree-one obstruction class.

(2) If $\delta^1 r = 0$ and there exists $b \in C^0$ with $\delta^0 b = r$, the residue is removable by a change of local representatives.

(3) If $\delta^1 r = 0$ and no such b exists, the residue represents a non-zero class in H^1.

Theorem 11.1 (Soundness of the finite residue classifier)**.** *Let*

$$C^0 \xrightarrow{\delta^0} C^1 \xrightarrow{\delta^1} C^2$$

be a finite cochain complex over $\mathbb{Q}$. For $r \in C^1$, the following are equivalent:

$$0 \neq [r] \in H^1$$

and

$$\delta^1 r = 0 \qquad \textit{and} \qquad r \notin \operatorname{im} \delta^0.$$

Proof. By definition,

$$H^1 = \ker \delta^1 / \operatorname{im} \delta^0.$$

The class $[r]$ is defined precisely when $r \in \ker \delta^1$, equivalently $\delta^1 r = 0$. That class is zero precisely when $r \in \operatorname{im} \delta^0$. Therefore $[r] \neq 0$ precisely when $\delta^1 r = 0$ and $r \notin \operatorname{im} \delta^0$. □

Lemma 11.2 (Cycle-pairing certificate)**.** *Let C_1 be the chain group dual to C^1, let $\partial : C_1 \to C_0$ be the boundary map dual to δ^0, and let $z \in C_1$ be a cycle, so $\partial z = 0$. If*

$$\langle z, r \rangle \neq 0,$$

then $r \notin \operatorname{im} \delta^0$.

Proof. Assume for contradiction that $r = \delta^0 b$ for some $b \in C^0$. Then

$$\langle z, r \rangle = \langle z, \delta^0 b \rangle = \langle \partial z, b \rangle = \langle 0, b \rangle = 0.$$

This contradicts $\langle z, r \rangle \neq 0$. Hence $r \notin \operatorname{im} \delta^0$. □

The cycle-pairing lemma is useful because it gives a small certificate. One does not need to trust a solver's failed linear-system report if one exhibits an explicit cycle whose pairing with the residue is non-zero.

12 The actual finite obstruction witness

We now give the finite object that has been checked computationally and proved by hand.

Let the finite nerve have four vertices

$$U_1, U_2, U_3, U_4$$

and four oriented edges

$$U_1U_2, \quad U_2U_3, \quad U_3U_4, \quad U_1U_4.$$

There are no two-simplices. Thus

$$C^0 = \mathbb{Q}^4, \qquad C^1 = \mathbb{Q}^4, \qquad C^2 = 0.$$

Consider the residue

$$r = (1, 1, 1, -2) \in C^1.$$

With the displayed ordering of vertices and edges, the coboundary matrix is

$$\delta^0 = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ -1 & 0 & 0 & 1 \end{pmatrix}.$$

For $b = (b_1, b_2, b_3, b_4)$, this says

$$\delta^0 b = (b_2 - b_1, b_3 - b_2, b_4 - b_3, b_4 - b_1).$$

Proposition 12.1 (Non-removability of the seam residue)**.** *The residue*

$$r = (1, 1, 1, -2)$$

determines a non-zero class in H^1*.*

Proof. Since $C^2 = 0$, we have $\delta^1 = 0$, hence

$$\delta^1 r = 0.$$

It remains to show that $r \notin \operatorname{im} \delta^0$. Suppose, for contradiction, that $r = \delta^0 b$ for some $b = (b_1, b_2, b_3, b_4)$. Then

$$b_2 - b_1 = 1, \qquad b_3 - b_2 = 1, \qquad b_4 - b_3 = 1,$$

so

$$b_4 - b_1 = (b_4 - b_3) + (b_3 - b_2) + (b_2 - b_1) = 3.$$

But the fourth component of $\delta^0 b = r$ requires

$$b_4 - b_1 = -2.$$

This is impossible. Therefore $r \notin \operatorname{im} \delta^0$. By Theorem 11.1,

$$0 \neq [r] \in H^1.$$

□

The same conclusion is certified by a cycle pairing. Let

$$z = (-1, -1, -1, 1) \in C_1.$$

A direct calculation gives

$$\partial z = 0.$$

The pairing with the residue is

$$\langle z, r \rangle = (-1)(1) + (-1)(1) + (-1)(1) + (1)(-2) = -5 \neq 0.$$

By Lemma 11.2, r is not a coboundary.

Corollary 12.2 (Two independent certificates)**.** *The non-zero class $[r] \in H^1$ is certified both by the inconsistent linear system $\delta^0 b = r$ and by the non-zero cycle pairing $\langle z, r \rangle = -5$.*

13 Presentation invariance and refinement witnesses

The finite obstruction should not depend on cosmetic presentation choices. It should also not disappear under the declared refinements. This section states exactly what is proved.

13.1 Presentation invariance

The following changes preserve the obstruction verdict.

(1) Region renaming.

(2) Edge orientation reversal, with the corresponding sign change.

(3) Non-zero rational scaling of the residue.

(4) Gauge perturbation by an exact coboundary.

(5) Edge ordering permutation.

The proof is standard. Region renaming, edge ordering, and orientation reversal are invertible changes of basis on the finite cochain complex. They induce isomorphic cohomology groups. Non-zero rational scaling preserves non-zeroness of the class. Gauge perturbation sends

$$r \longmapsto r + \delta^0 b,$$

and therefore

$$[r + \delta^0 b] = [r].$$

Proposition 13.1 (Declared presentation invariance)**.** *For the finite object of Section 12, the obstruction verdict*

$$0 \neq [r] \in H^1$$

is invariant under the five declared presentation transformations above.

Proof. The first, second, and fifth transformations are invertible changes of basis on C^0 and C^1 that commute with the displayed coboundary after transport of structure. Hence they preserve membership in $\ker \delta^1$ and $\operatorname{im} \delta^0$. Non-zero scaling multiplies $[r]$ by an element of $\mathbb{Q}^\times$, so it cannot turn a non-zero class into zero. Finally, exact gauge perturbation does not change the cohomology class at all. Therefore the obstruction verdict is preserved in each case. □

13.2 Refinement witnesses

Refinement persistence is a more delicate claim. The present paper proves persistence for the declared refinement witnesses. It does not claim persistence under all possible refinements.

Let

$$\rho^* : C^1(N; \mathbb{Q}) \to C^1(N'; \mathbb{Q})$$

be one of the declared transfer maps to a refined nerve, and set

$$r' = \rho^* r.$$

To prove $[r'] \neq 0$, it is enough to exhibit a refined cycle z' such that

$$\partial z' = 0 \qquad \text{and} \qquad \langle z', r' \rangle \neq 0.$$

This is exactly Lemma 11.2 applied in the refined complex.

The four declared refinement witnesses are as follows.

Refinement	Verdict	Pairing
Subdivide U_1	nontrivial H^1 obstruction	$-7/2$
Subdivide U_2	nontrivial H^1 obstruction	-4
Subdivide all regions	nontrivial H^1 obstruction	$-5/4$
Insert bridge between U_1 and U_2	nontrivial H^1 obstruction	-5

Each row is a self-contained obstruction certificate: a refined cycle exists and pairs non-trivially with the transferred residue. Therefore each transferred residue is not a coboundary.

Proposition 13.2 (Persistence under declared refinement witnesses)**.** *For each of the four declared refinements in the table, the transferred residue $r' = \rho^* r$ represents a non-zero class in the refined H^1.*

Proof. For each refinement, the refined certificate supplies a cycle z' with $\partial z' = 0$ and a non-zero rational pairing $\langle z', r' \rangle$, listed in the table. By Lemma 11.2, $r' \notin \operatorname{im} \delta'^0$. The refined residue is a cocycle in the corresponding refined complex. Hence $[r'] \neq 0$ in the refined H^1. □

Definition 13.3 (Cycle-faithful refinement)**.** *Let $z \in Z_1(N; \mathbb{Q})$. A refinement $\rho : N' \to N$ equipped with a chain pushforward $\rho_* : C_1(N'; \mathbb{Q}) \to C_1(N; \mathbb{Q})$ is* cycle-faithful relative to z *if*

$$\rho_*(Z_1(N'; \mathbb{Q})) \cap \mathbb{Q}^\times z \neq \varnothing.$$

Equivalently, there exists $z' \in Z_1(N'; \mathbb{Q})$ and $\lambda \in \mathbb{Q}^\times$ such that $\rho_* z' = \lambda z$.

Definition 13.4 (Nonzero-degree lift of a cycle)**.** *Let $z \in Z_1(N; \mathbb{Q})$ and let $\rho_* : C_1(N'; \mathbb{Q}) \to C_1(N; \mathbb{Q})$ be a chain pushforward. A cycle $z' \in Z_1(N'; \mathbb{Q})$ is a* nonzero-degree lift *of z if there exists $\lambda \in \mathbb{Q}^\times$ such that*

$$\rho_* z' = \lambda z.$$

The refinement ρ is cycle-faithful relative to z (Definition 13.3) precisely when a nonzero-degree lift of z exists.

Theorem 13.5 (Cycle-lift persistence)**.** *Let $r \in Z^1(N; \mathbb{Q})$, and suppose there exists $z \in Z_1(N; \mathbb{Q})$ such that $\langle z, r \rangle \neq 0$. Let $\rho : N' \to N$ be a refinement with adjoint maps*

$$\rho^* : C^1(N; \mathbb{Q}) \to C^1(N'; \mathbb{Q}), \qquad \rho_* : C_1(N'; \mathbb{Q}) \to C_1(N; \mathbb{Q})$$

satisfying $\langle z', \rho^* \alpha \rangle = \langle \rho_* z', \alpha \rangle$ for all z', α.

If z admits a nonzero-degree lift z' (Definition 13.4), then $[\rho^* r] \neq 0 \in H^1(N'; \mathbb{Q})$.

Proof. Assume, for contradiction, that $\rho^* r$ is a coboundary. Then there exists $b' \in C^0(N'; \mathbb{Q})$ such that

$$\rho^* r = \delta'^0 b'.$$

Since z' is a cycle, $\partial' z' = 0$. Hence

$$\langle z', \rho^* r \rangle = \langle z', \delta'^0 b' \rangle = \langle \partial' z', b' \rangle = 0.$$

But by adjointness and the cycle-lift condition,

$$\langle z', \rho^* r \rangle = \langle \rho_* z', r \rangle = \langle \lambda z, r \rangle = \lambda \langle z, r \rangle.$$

Since $\lambda \neq 0$ and $\langle z, r \rangle \neq 0$, this quantity is nonzero. Contradiction. Therefore $\rho^* r$ is not a coboundary, and so

$$[\rho^* r] \neq 0 \in H^1(N'; \mathbb{Q}).$$

□

Remark 13.6 (Which declared refinements satisfy the theorem hypotheses)**.** *The adjointness condition holds automatically for all four declared refinements because ρ_* is taken to be the transpose of ρ^*. The cycle-lift condition $\rho_* z' = \lambda z$ holds for two of the four:*

Refinement	*Cycle-lift*	λ	*Proof method*
Subdivide U_1	*No*	—	*Direct pairing*
Subdivide U_2	*No*	—	*Direct pairing*
Subdivide all regions	*Yes*	1/4	*Theorem 13.5*
Insert bridge U_1–U_2	*Yes*	1	*Theorem 13.5*

For the two single-region subdivisions, the cycle-lift condition is impossible under equal-distribution transfer: flow conservation at the junction vertex forces $\rho_*(z')[e_{\text{in}}] = \frac{1}{2}\rho_*(z')[e_{\text{out}}]$, which breaks proportionality with z unless $\lambda = 0$. Those two refinements are therefore certified by the direct cycle-pairing Lemma 11.2 applied in N', not by Theorem 13.5.

The distinction is verified computationally in `cycle_lift_test.py`*, which solves the homogeneous system*

$$\begin{pmatrix} \partial' & 0 \\ \rho_* & -z \end{pmatrix} \begin{pmatrix} z' \\ \lambda \end{pmatrix} = 0$$

over $\mathbb{Q}$ and proves algebraically that $\lambda = 0$ is forced for the two single-region subdivisions.

Proposition 13.7 (Rank criterion for cycle-faithfulness)**.** *Let B' be the boundary matrix of N', let P be the matrix of $\rho_* : C_1(N';\mathbb{Q}) \to C_1(N;\mathbb{Q})$, and let K be a matrix whose columns form a basis for $\ker B' = Z_1(N';\mathbb{Q})$. Then ρ is cycle-faithful relative to z if and only if*

$$\operatorname{rank}(PK) = \operatorname{rank}([PK \;\; z]).$$

Proof. Cycle-faithfulness holds iff $z \in \operatorname{im}(\rho_*|_{Z_1(N')})$. The columns of PK span $\operatorname{im}(\rho_*|_{Z_1(N')})$. By the standard rank test for membership in a column space over a field, $z \in \operatorname{span}(PK)$ iff appending z does not increase the rank of PK. □

Refinement	$\dim Z_1(N')$	$\operatorname{rank}(PK)$	$\operatorname{rank}([PK\, z])$	Cycle-faithful
Subdivide U_1	3	1	2	No
Subdivide U_2	3	1	2	No
Subdivide all regions	13	1	1	Yes
Insert bridge U_1–U_2	1	1	1	Yes

Remark 13.8 (Two layers of persistence)**.** *Refinement persistence has two layers.*

Direct obstruction persistence. *There exists a refined cycle $z' \in Z_1(N';\mathbb{Q})$ with $\langle z', \rho^* r\rangle \neq 0$. All four declared refinements satisfy this. Proved by Lemma 11.2 in N'.*

Witness persistence (cycle-faithfulness). *The original obstruction cycle z admits a nonzero-degree lift. Two of the four declared refinements satisfy this. When it holds, Theorem 13.5 applies without computing a new cycle. An obstruction can persist even when its original witness-cycle does not lift.*

Conjecture 13.9 (Balanced cycle-fibre)**.** *Let z be a simple rational cycle in a finite oriented nerve N. Let $\rho : N' \to N$ be a refinement equipped with a chain pushforward ρ_*. Suppose the full preimage of the support of z contains a divergence-free rational circulation whose pushforward crosses every edge of z with the same nonzero rational degree. Then ρ is cycle-faithful relative to z.*

Remark 13.10 (Evidence and rank interpretation)**.** *For the insert-bridge and subdivide-all refinements, such a uniform circulation exists: the bridge cycle traverses both halves of the split edge with the same weight, and the diagonal path through the fully subdivided complex crosses each cluster with weight* $1/4$*. In both cases* $\operatorname{rank}(PK) = \operatorname{rank}([PK \;\; z]) = 1$*.*

For the single-region subdivisions, no uniform circulation exists. The generator of $\operatorname{im}(P|_{Z_1(N')})$ *has a non-uniform ratio relative to* z*. For subdivide-*U_1*:*

$$\textit{generator} = (-\tfrac{1}{2}, -1, -1, \tfrac{1}{2}), \quad \frac{\textit{gen}[e]}{z[e]} = \begin{cases} \frac{1}{2} & e \in \{U_1U_2, U_1U_4\} \\ 1 & e \in \{U_2U_3, U_3U_4\} \end{cases}.$$

The split edges (those incident to the subdivided U_1*) contribute ratio* $\frac{1}{2}$*; the unchanged edges contribute ratio* 1*. No scalar multiple of this generator equals* z*, so* $\operatorname{rank}([PK \;\; z]) = 2$ *and* $z \notin \operatorname{im}(PK)$*.*

Proving Conjecture 13.9 would give a geometric sufficient condition for cycle-faithfulness. The two theorems below provide partial progress.

Definition 13.11 (Cycle-covering refinement)**.** *Let* $z \in Z_1(N;\mathbb{Q})$ *be a simple rational cycle, and let* $L = \operatorname{supp}(z)$ *be its supporting loop. A refinement* $\rho : N' \to N$ *is* cycle-covering relative to z *if there exists a subcomplex* $L' \subseteq N'$ *and a rational cycle* $z' \in Z_1(L';\mathbb{Q})$ *such that* $\rho_* z' = \lambda z$ *for some* $\lambda \in \mathbb{Q}^\times$*.*

Theorem 13.12 (Cycle-covering refinements are cycle-faithful)**.** *Every cycle-covering refinement relative to* z *is cycle-faithful relative to* z*.*

Proof. By definition, a cycle-covering refinement supplies $z' \in Z_1(N';\mathbb{Q})$ with $\rho_* z' = \lambda z$ and $\lambda \neq 0$. Hence $\rho_*(Z_1(N';\mathbb{Q})) \cap \mathbb{Q}^\times z \neq \varnothing$, which is the definition of cycle-faithfulness. □

Theorem 13.13 (Uniform loop subdivision)**.** *Let* z *be a simple rational cycle in* N*. Suppose* $\rho : N' \to N$ *refines each edge* e *of* z *by replacing it with a nonempty oriented path* $\Pi(e)$ *in* N'*, and that the replacement paths chain consecutively to form a closed loop* z' *in* N'*. If there exists* $\lambda \in \mathbb{Q}^\times$ *such that* $\rho_*(z')[e] = \lambda z(e)$ *for every edge* e *in* z*, then* ρ *is cycle-faithful relative to* z*.*

Proof. The paths chain into a closed loop by hypothesis, so $\partial' z' = 0$. The pushforward condition gives $\rho_* z' = \lambda z$ by summing the per-edge contributions. Since $\lambda \neq 0$, z' is a nonzero-degree lift of z, and Theorem 13.5 applies. □

Remark 13.14 (Application to the declared refinements)**.** *For* insert bridge*: the edge* U_1U_2 *is replaced by the serial path* U_1*-Bridge-*U_2*, and* z' *traverses both path edges. The pushforward gives degree* $\lambda = 1$ *on every edge.*

For subdivide all*: the diagonal path* $U_{1a}U_{2a}U_{3a}U_{4a}U_{1a}$ *replaces each base edge with one of four parallel options. Each contributes degree* $\frac{1}{4}$*, giving* $\lambda = \frac{1}{4}$ *uniformly.*

For subdivide U_1 or U_2*: the edges incident to the subdivided region contribute degree* $\frac{1}{2}$ *while the unchanged edges contribute degree* 1*. The degree is non-uniform, so Theorem 13.13 does not apply. Non-faithfulness is proved algebraically by* $\operatorname{rank}([PK \;\; z]) = 2$*.*

The full picture is summarised in the following table.

Refinement	Direct persistence	Cycle-faithful	Degree λ	Reason
Subdivide U_1	Yes	No	—	cycle image line misses z
Subdivide U_2	Yes	No	—	cycle image line misses z
Subdivide all	Yes	Yes	1/4	uniform diagonal loop
Insert bridge	Yes	Yes	1	preserved serial loop

14 Spectral diagnosis of associator fields

Cohomology classifies closed defect fields up to coboundary. Before closure is known, one may still analyse the defect field spectrally. This section gives a finite-dimensional diagnostic version.

Assume that the nerve $N(\mathcal{U})$ is oriented and that the realised associator field has scalar values in $\mathbb{R}$, or more generally values in finite-dimensional real inner-product spaces with an induced inner product on cochains. Then the simplicial cochain spaces $C^n(N;\mathbb{R})$ carry coboundary maps

$$d_n : C^n \to C^{n+1}$$

and adjoints

$$d_n^* : C^{n+1} \to C^n.$$

The degree-two Hodge Laplacian is

$$\Delta_2 = d_1 d_1^* + d_2^* d_2.$$

Equip each cochain space $C^n(N;\mathbb{R})$ with the standard inner product

$$\langle \phi, \psi \rangle = \sum_J \phi_J \psi_J,$$

summed over the oriented n-simplices J of the nerve.

Theorem 14.1 (Spectral decomposition of the associator field)**.** *Assume the nerve $N(\mathcal{U})$ is oriented and that the realised associator field is scalar,*

$$\mathcal{A} \in C^2(N;\mathbb{R}).$$

Then $\mathcal{A}$ admits a unique orthogonal decomposition

$$\mathcal{A} = d_1\theta + H + d_2^*\eta,$$

where

(i) $d_1\theta \in \mathrm{im}(d_1)$ is the exact component;

(ii) $H \in \ker(\Delta_2)$ is the harmonic component;

(iii) $d_2^*\eta \in \mathrm{im}(d_2^*)$ is the coexact component.

If, in addition, the scalar realisation identifies admissible pairwise repair with d_1, then the exact component is repairable by pairwise gauge change. If fourfold coherence is tested by d_2, then a non-zero coexact component records failure of fourfold coherence. The harmonic component is the part left by both tests.

Proof. The degree-two Hodge Laplacian is $\Delta_2 = d_1 d_1^* + d_2^* d_2$. Because $C^2(N;\mathbb{R})$ is a finite-dimensional inner-product space, finite Hodge theory yields the orthogonal direct sum

$$C^2 = \mathrm{im}(d_1) \oplus \ker(\Delta_2) \oplus \mathrm{im}(d_2^*).$$

Projecting $\mathcal{A}$ onto the three summands gives the stated decomposition. Orthogonality makes it unique. The identification of these subspaces with operational behaviour is conditional. When the chosen scalar realisation identifies the boundary gauge variation operator D with d_1, the subspace $\mathrm{im}(d_1)$ records the repairable directions. When fourfold coherence is tested by d_2, the subspace $\mathrm{im}(d_2^*)$ records the residual failure of d_2-closure. The harmonic component is then the part annihilated by both tests. $\square$

Remark 14.2 (Fourier language). *The eigenvectors of Δ_2 play the role of finite Fourier modes on the gluing nerve. Expanding the associator field in these modes reveals whether the defect is concentrated on isolated seams, spread across a global pattern, or carried by harmonic modes. This is a diagnostic use of spectral theory, not a replacement for the algebraic localisation theorem.*

15 A non-vanishing finite pattern

The following example shows that the indexing geometry of triple seams can support non-zero two-dimensional obstruction data.

Example 15.1 (Tetrahedral boundary nerve). *Let $\mathcal{U} = (U_1, U_2, U_3, U_4)$ have nerve equal to the boundary of a tetrahedron. Thus all pairwise and triple overlaps are non-empty, while the fourfold overlap is empty. With constant coefficients in a field k, the nerve is a triangulated 2-sphere. Hence*

$$H^2(N(\mathcal{U}); k) \cong k.$$

Choose a 2-cochain A whose evaluation on the oriented fundamental 2-cycle is non-zero. Since there are no 3-simplices, every 2-cochain is closed. Since A pairs non-trivially with the fundamental class, it is not a coboundary. It therefore represents a non-zero coefficient-level obstruction class.

Remark 15.2 (Coefficient-level example). *The example shows that the obstruction group for the indexing nerve can be non-zero. It does not by itself prove that every non-zero class is realised by a boundary-corrected multiplication. Realisation is an additional algebraic question.*

16 Computational artefact and reproducibility

The finite witness of Sections 11 and 12 is implemented as an exact rational classifier. The implementation has five roles.

(1) It builds the finite coboundary matrices δ^0 and δ^1.

(2) It checks whether the residue is a cocycle.

(3) It solves the linear system $\delta^0 b = r$ over $\mathbb{Q}$.

(4) It emits a certificate containing the cohomology summary and cycle witness.

(5) It reruns presentation, refinement, and property-based regression tests.

The property-based regression test runs 1000 random rational residues on the four-cycle and checks the classifier against the cycle-pairing criterion:

$$\langle z, r \rangle = 0 \quad \Longleftrightarrow \quad r \text{ is removable,}$$

and

$$\langle z, r \rangle \neq 0 \quad \Longleftrightarrow \quad r \text{ represents a nontrivial } H^1 \text{ obstruction.}$$

The current repository state records 1000 passed cases and zero failed cases.

This computational artefact is not used as a substitute for the proof. The proof of the actual residue is the explicit contradiction and the cycle pairing. The code provides reproducibility, regression protection, and a way of generating further finite witnesses.

17 Interpretation

The central lesson is simple:

pairwise interfaces do not exhaust the structure of gluing.

A local correctness proof may be valid on every region U_i. It may also be valid on every pairwise overlap $U_i \wedge U_j$. The global claim can still fail at $U_i \wedge U_j \wedge U_k$, because the two histories of pairwise composition need not leave the same residue.

This is why the associator field is a useful primitive object. It does not merely announce failure. It says where the failure lives, which triples participate in it, and whether it has the form of an isolated seam residue, a repairable bookkeeping artefact, or a persistent global mode.

In distributed systems, one may read regions as subsystems and overlaps as shared interfaces. The corrected product models reconciliation across an interface. The associator field then marks three-way integration failures that are not certified away by unit tests and pairwise interface tests alone.

In verification, the same structure distinguishes local proof from compositional proof. A module may satisfy its local specification. A pair of modules may satisfy an interface contract. Three modules may still fail to compose coherently. The associator field is the finite witness of that failure.

The finite H^1 example adds a second lesson. Once a residue has been found, the correct question is not whether it is non-zero. The correct question is whether it is removable. A residue that is exact is a bookkeeping artefact. A residue that is closed and not exact is a genuine obstruction relative to the declared finite apparatus.

The lesson is not that all verification problems are cohomological. That would be too strong. The lesson is that some local-to-global failures have a finite boundary structure, and that associator fields and residue classifiers provide disciplined ways of seeing that structure before choosing a classification method.

18 Limitations and next steps

The limitations are part of the construction. They mark where the current finite theory is exact, where classification requires extra structure, and where the next mathematical problems begin.

18.1 Current methodological scope

First, the support-refinement axiom is strong. It isolates seam-supported multiplication rather than arbitrary regional multiplication. This makes Theorem 5.3 clean, but narrows the class of examples.

Second, the first-order variation formula requires a genuine first-order boundary regime. Correction and gauge maps must not reintroduce higher-order dependence on already-boundary terms. This is a guardrail, not a flaw. It is what makes the variation operator linear enough to be written as the four-term expression of Section 8.

Third, spectral Hodge analysis requires scalar or finite-dimensional inner-product values. It is a diagnostic layer, not part of the basic algebraic construction.

Fourth, the finite residue classifier proves a degree-one obstruction for a declared finite object. It does not by itself classify every associator field, nor does it prove that every triple-seam associator field reduces to a degree-one residue. The two constructions are related by the repair problem, but they are not identical.

18.2 Classification limits

The repair quotient $\mathrm{Field}^2 / \operatorname{im} D$ depends on the chosen class of admissible boundary gauges. Different repair regimes may produce different obstruction notions. This dependence is unavoidable, because repair is not a purely formal declaration. It is constrained by what pairwise boundary changes are allowed.

Čech cohomology is also not automatic. It requires a coefficient system with restriction maps and a fourfold coherence condition. The paper deliberately keeps these structures separate from the associator field itself. Otherwise one would confuse the finite defect with one possible classification of that defect.

The refinement result in Section 13 is deliberately scoped. It proves persistence for the four declared refinement witnesses. A universal refinement theorem requires a declared admissible refinement class and a proof of the pairing-preservation identity

$$\langle z', \rho^* r\rangle = \langle \rho_* z', r\rangle.$$

18.3 Horizons for generalisation

The associator field is input-dependent. For a fixed input family a, one obtains $\mathcal{A}_a$. A more invariant theory would treat associators as operator-valued fields. That would replace a static residue with a transformation-level diagnostic, capable of tracking how a seam defect acts on families of inputs rather than on one selected input family.

A second horizon is realisability. The theory should identify which finite triple-indexed fields can be produced by some supported regional algebra and some admissible boundary correction. Such a theorem would separate accidental examples from structurally forced patterns.

A third horizon is dynamic repair. In distributed systems, the observation boundary may shift over time: sensors degrade, network partitions open and close, and interfaces appear or disappear. The present paper treats the regional cover as fixed. A dynamic version would have to track associator fields over changing nerves and changing admissible repair operators.

A fourth horizon is formal verification. The executable classifier can be moved from tested Python to a verified kernel. A minimal formal theorem would state that if the checker returns `nontrivial_H1_obstruction`, then the supplied residue represents a non-zero H^1 class. That theorem belongs in a proof assistant such as Rocq, Lean, or Agda. The executable artefact can compute certificates; the proof assistant can certify the classifier's soundness.

These are not afterthoughts. They are the next tasks opened by the present construction: realisability, operator-valued associator fields, dynamic repair, universal refinement persistence, and formal verification of the finite classifier.

19 Conclusion

Gluing is not exhausted by local validity and pairwise compatibility. A finite regional system may be locally associative on every region and explicitly corrected on every pairwise seam while still producing a parenthesisation residue on a triple seam. The associator field records that residue before it is forced into any particular classification scheme.

The paper has proved the basic localisation theorem: under support-refining boundary corrections, the associator defect of a corrected triple product is supported on the common triple overlap. It has then separated detection, repair, and obstruction. Cohomology and spectral analysis enter as structured classifications of the field, not as assumptions built into the field from the start.

Finally, the paper has given an exact finite obstruction witness. The residue $r = (1, 1, 1, -2)$ on a four-region seam cycle is a cocycle but not a coboundary. It is certified by an explicit contradiction and by the cycle pairing $\langle z, r\rangle = -5$. Its verdict is invariant under the declared presentation changes and persists under the four declared refinement witnesses.

The resulting claim is modest but sharp. The defect is not merely a non-zero value. It is a non-zero cohomology class relative to the declared finite regional apparatus. That is the point at which a seam residue becomes an obstruction.

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